Math 1232 Summer 2026
Practice Final

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, **two-sided** cheat sheet you have made for yourself ahead of time.
- You may use a four-function calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- Each problem is worth ten points. The whole test is scored out of 100 points.
- The exam has 14 problems, the first 8 of which you must do.
- You must solve at least 2 of the last 6 problems. If you solve more than 2, your best 2 scores will be recorded for points as problems 9 and 10, and any additional (successful) solutions will count for a bonus point or two.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

1	2	3	4	5
6	7	8	9	10

Total ____ /100

Name:

Name: ______________________________

Problem 1. Find the general solution to the equation $y' = e^{x+y}$.

Problem 2. Let $b_n = \dfrac{n! - 2}{(n+2)!}$. Compute $\lim_{n\to\infty} b_n$ with justification.

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Problem 3. Compute $\displaystyle\sum_{n=0}^{\infty} \ln\left(\frac{n+1}{n+3}\right)$.

Problem 4. Analyze the convergence of $\displaystyle\sum_{n=2}^{\infty} \frac{3n^3 - 2n^2 + n}{4n^6 + 4n^3 - n}$.

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Problem 5. Analyze the convergence of $\displaystyle\sum_{n=2}^{\infty}(-1)^n\frac{n^3-n}{3n^4+n^2+2}$.

Problem 6. Find the radius of convergence and the interval of convergence of $\displaystyle\sum_{n=0}^{\infty}\frac{4^n}{n^2+n}x^n$.

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Problem 7. Write down a formula for the Taylor series of $(1-2x)^{-3}$, and then write down the third degree polynomial $T_3(x)$ explicitly.

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Problem 8. Do one of the following integrals:

$$\text{(a)} \int \frac{3x^2 - 7x + 8}{x(x-2)^2}\,dx, \qquad \text{(b)} \int x^3\sqrt{1-x^2}\,dx, \qquad \text{(c)} \int \arcsin(x)\,dx.$$

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Do at least 2 of the following 6 problems. If you successfully solve more than 2, you will receive one or two bonus points on the exam, but you should make sure to have at least something written down for each of the first 8 problems before attempting more than 2 problems in this section.

Problem 9. Is $f(x) = e^{x^3+1}$ invertible or not? Justify your answer. If it's invertible, give a formula for the inverse.

Problem 10. Compute $\lim_{x\to 0} \dfrac{7^x - 5^x}{x}$.

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Problem 11. Compute $\displaystyle\int_0^7 \frac{1}{\sqrt[3]{7-x}}\,dx$.

Problem 12. Compute the arc length of the curve $y = \frac{1}{3}(x^2-2)^{3/2}$ for $3 \leq x \leq 6$.

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Problem 13. Use a Taylor series to compute $\lim_{x \to 0} \dfrac{\cos(x^2) - 1 + x^4/2}{x^8}$.

Problem 14. Find a **clockwise** parameterization for an ellipse 10 units wide and 6 units tall (thus with major radius 5 and minor radius 3), centered at the point $(5, 2)$.